Calcurion Physics Worksheets

A-Level Physics

Topic: Quantum Physics

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. State the photoelectric equation. **[2]**

2. Define the work function of a metal. **[1]**

3. Light of frequency 8.0×10^{14} Hz is incident on a metal with work function 2.0 eV. Calculate the maximum KE of photoelectrons. ($h = 6.63 \times 10^{-34}$) **[4]**

4. What is the threshold frequency? **[1]**

5. Calculate the threshold frequency for a metal with $\phi = 1.8$ eV. **[3]**

6. State de Broglie's relation. **[1]**

7. Calculate the de Broglie wavelength of an electron moving at 1.0×10^6 m/s. ($m_e = 9.1 \times 10^{-31}$) **[3]**

8. What is an energy level in an atom? **[1]**

9. An electron transitions from $n = 3$ to $n = 1$ in hydrogen. Energy levels are -1.5 eV and -13.6 eV. Calculate the photon energy. **[2]**

10. Calculate the wavelength of that photon. **[3]**

11. State the ground state of an atom. **[1]**

12. What is ionisation energy? **[1]**

13. Explain why only certain wavelengths appear in an emission spectrum. **[3]**

14. What is wave-particle duality? **[2]**

15. Name an experiment that demonstrates the wave nature of electrons. **[1]**

Section B – Hard (Q16–25)

16. Explain why the photoelectric effect supports the particle nature of light. **[3]**

17. UV light of wavelength 250 nm is incident on a metal with $\phi = 3.0$ eV. Calculate the stopping potential. **[4]**

18. An electron is accelerated through 50 V. Calculate its de Broglie wavelength. **[4]**

19. Sketch the energy level diagram for hydrogen showing the Lyman and Balmer series. **[3]**

20. Calculate the frequency of radiation emitted when an electron falls from $n = 4$ to $n = 2$ in hydrogen ($E_4 = -0.85$ eV, $E_2 = -3.4$ eV). **[3]**

21. Explain the difference between absorption and emission spectra. **[3]**

22. A photon has energy 2.5 eV. Calculate its momentum. **[3]**

23. Why does increasing light intensity not increase the maximum KE of photoelectrons? **[2]**

24. Describe electron diffraction and what it demonstrates. **[3]**

25. Calculate the minimum wavelength of X-rays produced when electrons accelerated through 25 kV strike a target. **[3]**

End of Worksheet • *Show all working clearly.*